

day 16 prep

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In [3]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

```
In [4]: def back(function, x_value, stepSize):
        return (function(x_value) - function(x_value - stepSize)) / stepSize

def forward(function, x_value, stepSize):
    return (function(x_value + stepSize) - function(x_value)) / stepSize

def mid(function, x_value, stepSize):
    return (function(x_value + stepSize) - function(x_value - stepSize)) / (2 * stepSize)

def doubled(function, x_value, stepSize):
    return (function(x_value + stepSize) - 2 * function(x_value) + function(x_value -
```

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In [12]: def f(x):
        return np.sin(x)
```

```
In [14]: x = np.linspace(0, 2 * np.pi, 100)
dx = x[1] - x[0]

y = f(x)

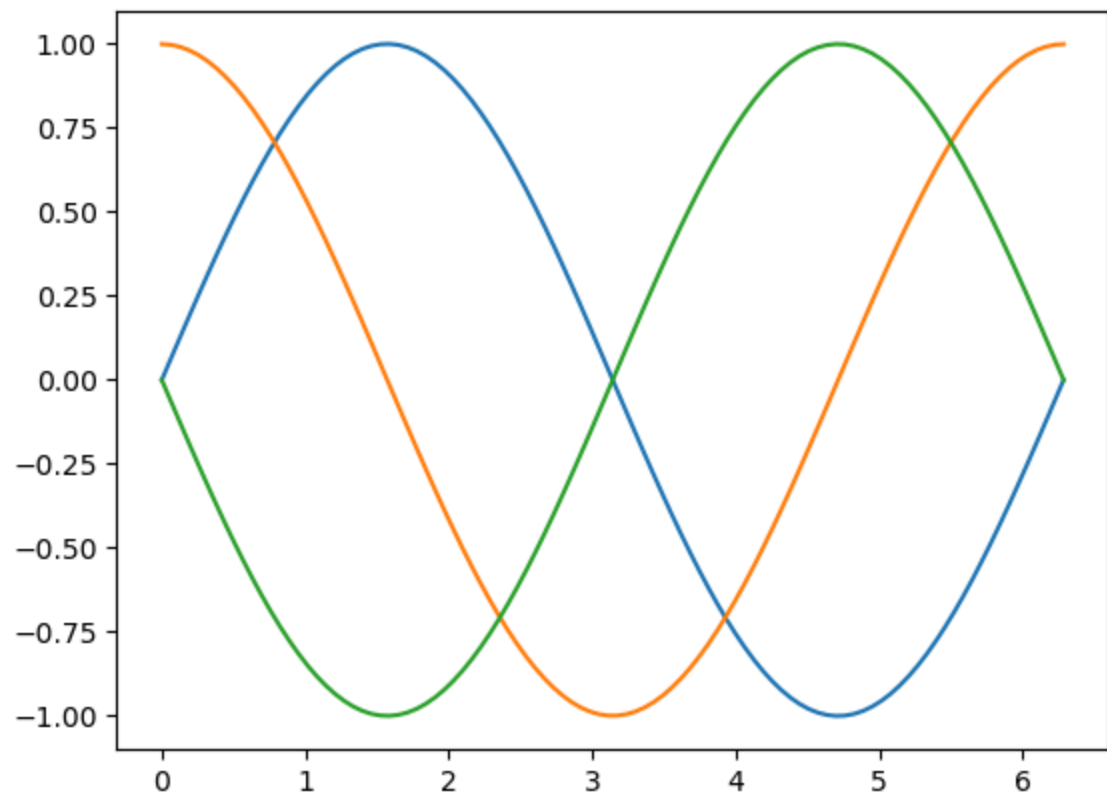
dy = [mid(f, x_, dx) for x_ in x]

ddy = [doubled(f, x_, dx) for x_ in x]

fig, ax = plt.subplots()

ax.plot(x, y)
ax.plot(x, dy)
ax.plot(x, ddy)
```

```
Out[14]: [<matplotlib.lines.Line2D at 0x7f8c0bed8c20>]
```



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In [ ]: np.gradient(
```

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In [ ]:
```